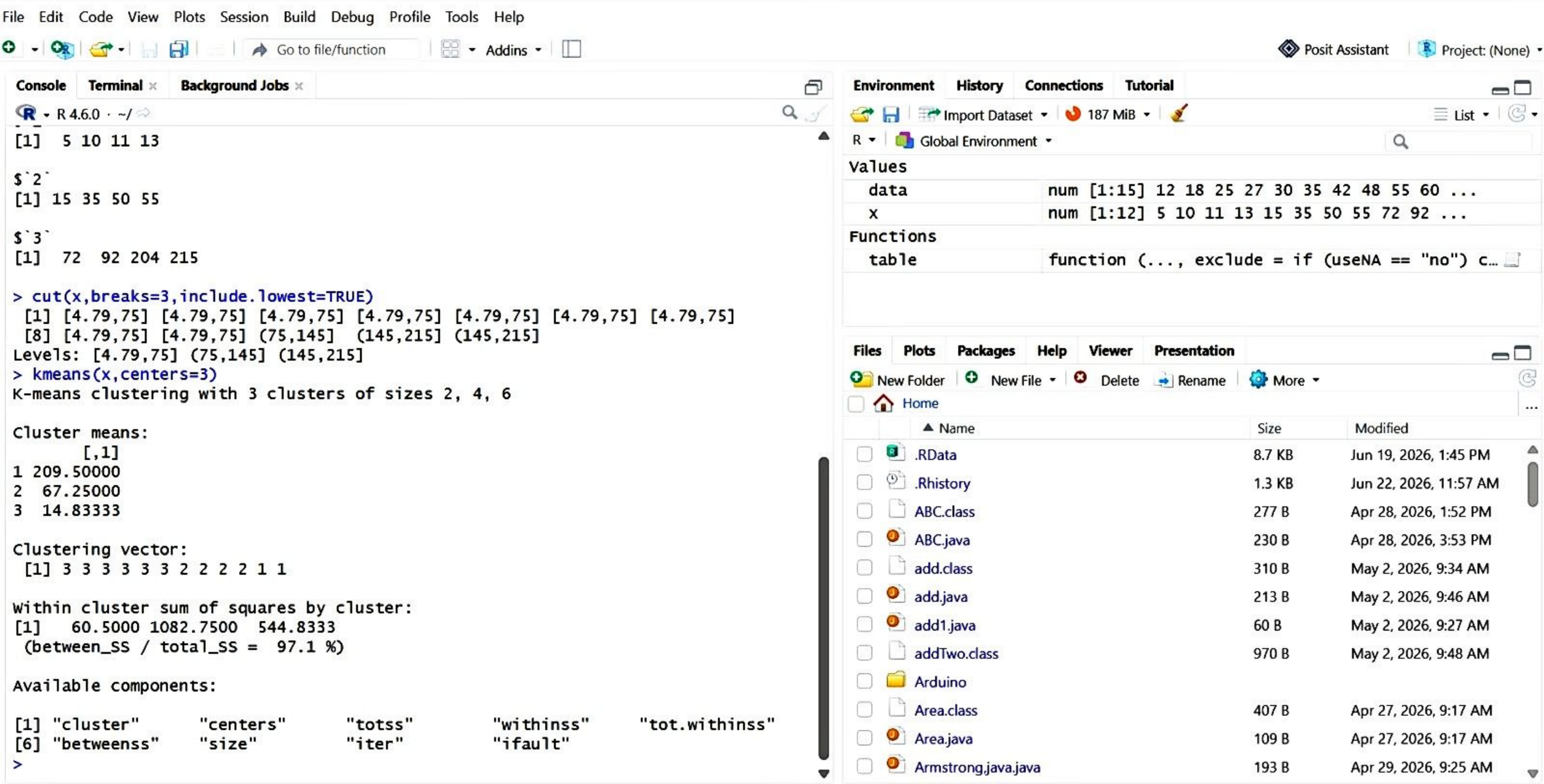
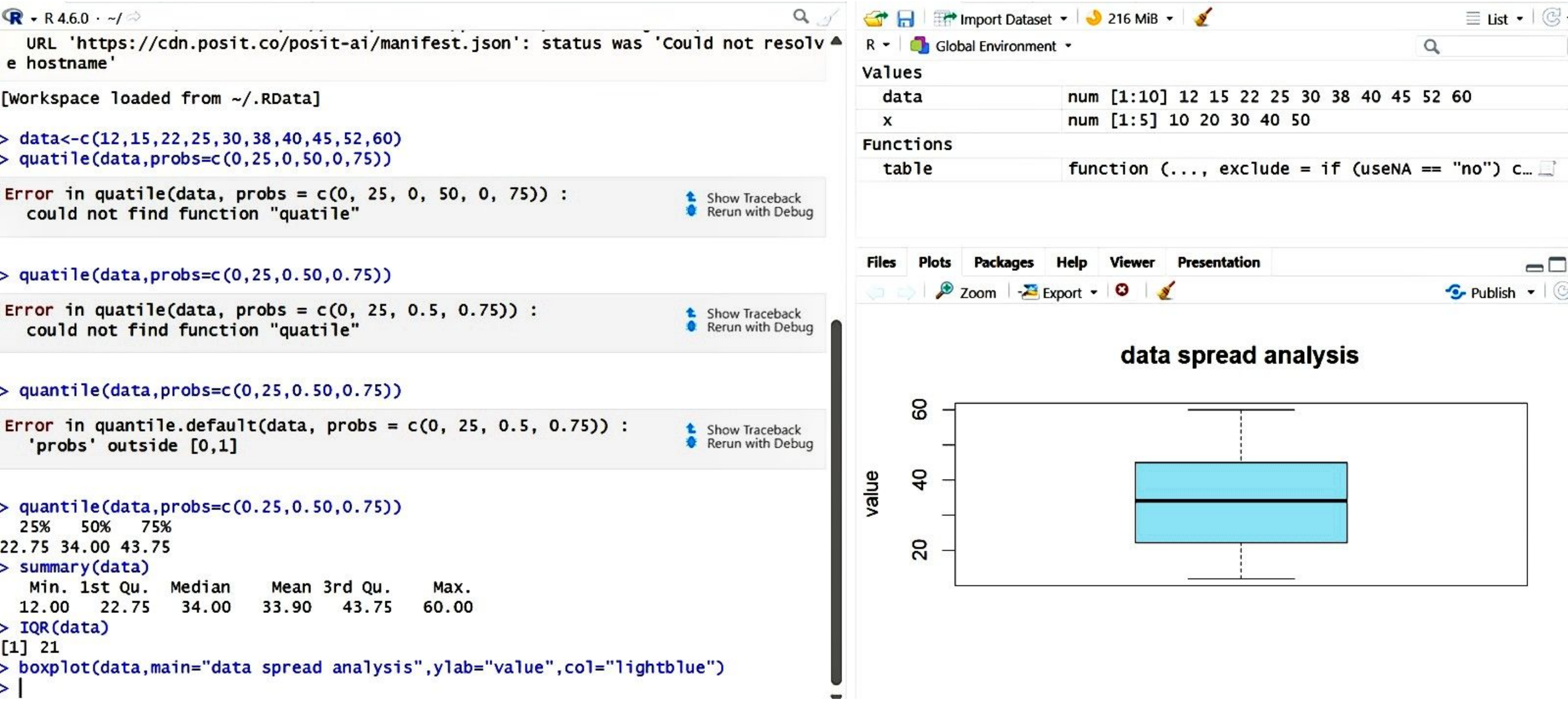

File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Posit Assistant
Project: (None)
Console
Terminal
Background Jobs
R 4.6.0 · ~/
[1] 5 10 11 13
$`2`
[1] 15 35 50 55
$`3`
[1] 72 92 204 215
> cut(x,breaks=3,include.lowest=TRUE)
[1] [4.79,75] [4.79,75] [4.79,75] [4.79,75] [4.79,75] [4.79,75] [4.79,75]
[8] [4.79,75] [4.79,75] (75,145] (145,215] (145,215]
Levels: [4.79,75] (75,145] (145,215]
> kmeans(x,centers=3)
K-means clustering with 3 clusters of sizes 2, 4, 6
Cluster means:
[,1]
1 209.50000
2 67.25000
3 14.83333
Clustering vector:
[1] 3 3 3 3 3 3 2 2 2 2 1 1
Within cluster sum of squares by cluster:
[1]   60.5000 1082.7500 544.8333
(between_SS / total_SS = 97.1 %)
Available components:
[1] "cluster"   "centers"   "totss"   "withinss"   "tot.withinss"
[6] "betweenss"   "size"   "iter"   "ifault"
>
Environment
History
Connections
Tutorial
Import Dataset
187 MiB
List
R
Global Environment
Values
data   num [1:15] 12 18 25 27 30 35 42 48 55 60 ...
x   num [1:12] 5 10 11 13 15 35 50 55 72 92 ...
Functions
table   function (..., exclude = if (useNA == "no") c...
Files
Plots
Packages
Help
Viewer
Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
.RData   8.7 KB   Jun 19, 2026, 1:45 PM
.Rhistory   1.3 KB   Jun 22, 2026, 11:57 AM
ABC.class   277 B   Apr 28, 2026, 1:52 PM
ABC.java   230 B   Apr 28, 2026, 3:53 PM
add.class   310 B   May 2, 2026, 9:34 AM
add.java   213 B   May 2, 2026, 9:46 AM
add1.java   60 B   May 2, 2026, 9:27 AM
addTwo.class   970 B   May 2, 2026, 9:48 AM
Arduino
Area.class   407 B   Apr 27, 2026, 9:17 AM
Area.java   109 B   Apr 27, 2026, 9:17 AM
Armstrong.java.java   193 B   Apr 29, 2026, 9:25 AM

```
> x <- c(10,20,30,40,50)
> mean(x)     #mean
[1] 30
> median(x)     #median
[1] 30
> #mode
> table(x)
x
10 20 30 40 50 
 1  1  1  1  1 
> #midrange
> (min(x)+ max(x))/2
[1] 30
> #quartiles
> quantile(x)
  0%  25%  50%  75% 100% 
  10   20   30   40   50 
> 
```

```
> x <- c(10,20,30,40,50)
> #min-max
> (x-min(x))/(max(x)- min(x))
[1] 0.00 0.25 0.50 0.75 1.00
> #z-score
> (x-mean(x))/sd(x)
[1] -1.2649111 -0.6324555  0.0000000  0.6324555  1.2649111
> |
```

```
> x <- c(10,20,30,40,50)
> #equal width binning
> cut(x,2)
[1] (9.96,30] (9.96,30] (9.96,30] (30,50]   (30,50]
Levels: (9.96,30] (30,50]
> #equal frequency binnig
> cut (x, quantile(x), include.loweat = TRUE)
[1] <NA>    (10,20] (20,30] (30,40] (40,50]
Levels: (10,20] (20,30] (30,40] (40,50]
> cut (x, quantile(x), include.lowest = TRUE)
[1] [10,20] [10,20] (20,30] (30,40] (40,50]
Levels: [10,20] (20,30] (30,40] (40,50]
>
```


R 4.6.0 · ~/
URL 'https://cdn.posit.co/posit-ai/manifest.json': status was 'Could not resolv
e hostname'
[Workspace loaded from ~/.RData]
> data<-c(12,15,22,25,30,38,40,45,52,60)
> quatile(data,probs=c(0,25,0,50,0,75))
Error in quatile(data, probs = c(0, 25, 0, 50, 0, 75)) :
could not find function "quatile"
Show Traceback
Rerun with Debug
> quatile(data,probs=c(0,25,0.50,0.75))
Error in quatile(data, probs = c(0, 25, 0.5, 0.75)) :
could not find function "quatile"
Show Traceback
Rerun with Debug
> quantile(data,probs=c(0,25,0.50,0.75))
Error in quantile.default(data, probs = c(0, 25, 0.5, 0.75)) :
'probs' outside [0,1]
Show Traceback
Rerun with Debug
> quantile(data,probs=c(0.25,0.50,0.75))
25% 50% 75%
22.75 34.00 43.75
> summary(data)
Min. 1st Qu. Median Mean 3rd Qu. Max.
12.00 22.75 34.00 33.90 43.75 60.00
> IQR(data)
[1] 21
> boxplot(data,main="data spread analysis",ylab="value",col="lightblue")
>
Import Dataset
216 MiB
List
R
Global Environment
Values
data num [1:10] 12 15 22 25 30 38 40 45 52 60
x num [1:5] 10 20 30 40 50
Functions
table function (..., exclude = if (useNA == "no") c…
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
data spread analysis
value
20
40
60

```
> data<-c(12,15,22,25,30,38,40,45,52,60)
> quantile(data,probs=c(0.25,0.50,0.75))
  25%   50%   75% 
22.75 34.00 43.75 
> summary(data)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  12.00   22.75   34.00   33.90   43.75   60.00 
> IQR(data)
[1] 21
> boxplot(data,main="data spread analysis", ylab="value",col="lightblue")
> 
```

```
>
>
>
> data<-c(12,18,25,27,30,35,42,48,55,60,75,90,120,150,200)
> cut(data,breaks=3)
 [1] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7]
 [7] (11.8,74.7] (11.8,74.7] (11.8,74.7] (11.8,74.7] (74.7,137]  (74.7,137]
[13] (74.7,137]  (137,200]   (137,200]
Levels: (11.8,74.7] (74.7,137] (137,200]
> quantile(data,prob=seq(0,1,length=4))
       0% 33.33333% 66.66667%      100%
 12.00000  33.33333  65.00000 200.00000
> kmeans(data,centers=3)
K-means clustering with 3 clusters of sizes 3, 2, 10

Cluster means:
   [,1]
1  95.0
2 175.0
3  35.2

Clustering vector:
 [1] 3 3 3 3 3 3 3 3 3 3 1 1 1 2 2

Within cluster sum of squares by cluster:
[1] 1050.0 1250.0 2249.6
 (between_SS / total_SS =  88.7 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"
> |
```

R · Global Environment

Values

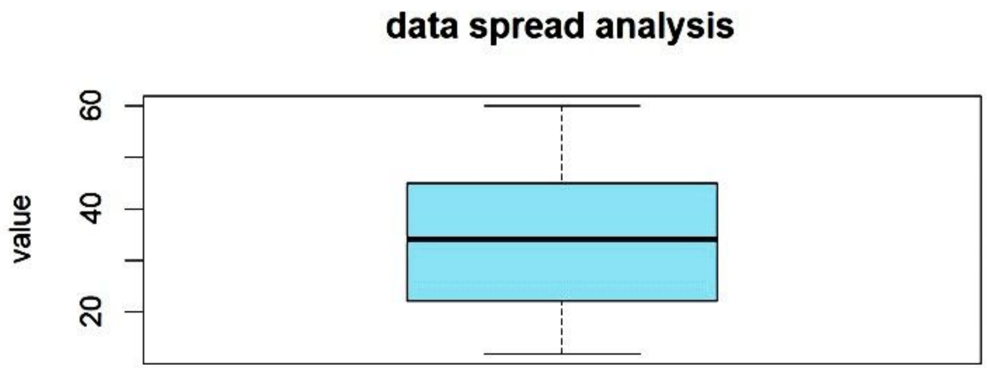
data	num [1:15] 12 18 25 27 30 35 42 48 55 60 ...
x	num [1:5] 10 20 30 40 50

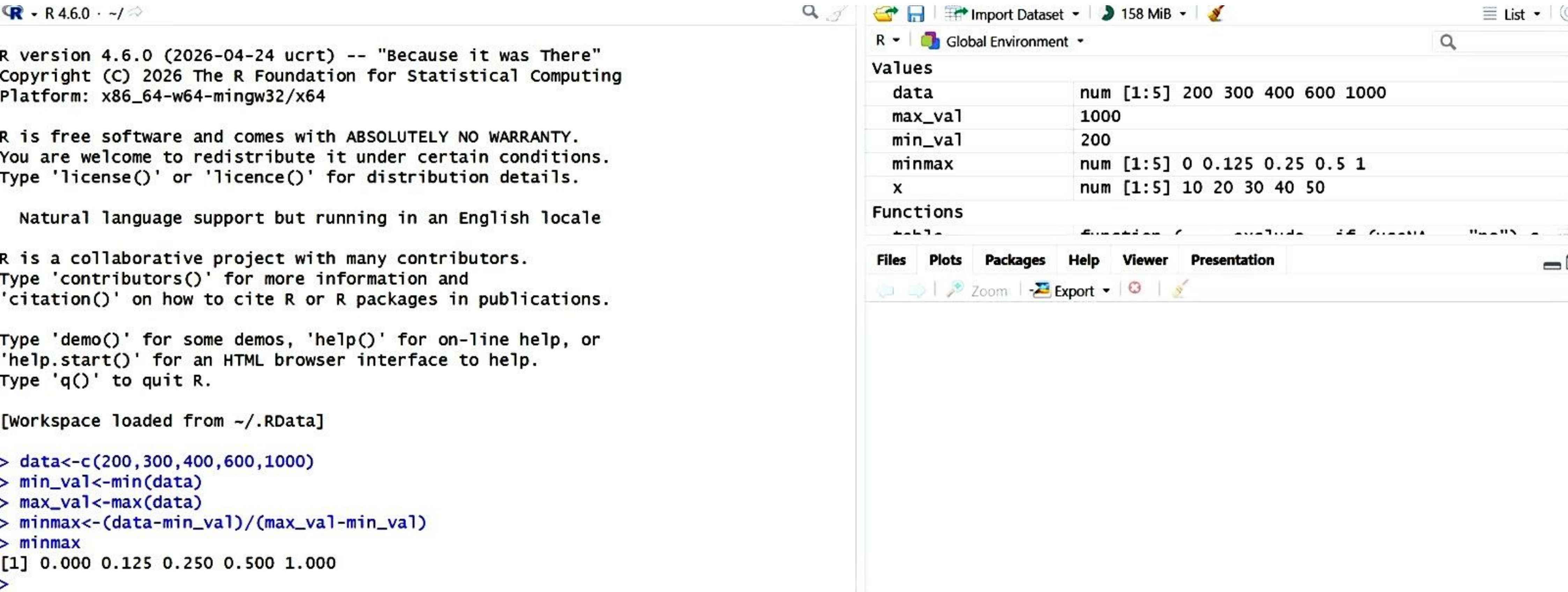
Functions

table	function (..., exclude = if (useNA == "no") c…

Files · Plots · Packages · Help · Viewer · Presentation

Zoom · Export · Publish




R 4.6.0 · ~/
R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
Natural language support but running in an English locale
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[Workspace loaded from ~/.RData]
> data<-c(200,300,400,600,1000)
> min_val<-min(data)
> max_val<-max(data)
> minmax<-(data-min_val)/(max_val-min_val)
> minmax
[1] 0.000 0.125 0.250 0.500 1.000
>
Import Dataset
158 MiB
List
R
Global Environment
Values
data
num [1:5] 200 300 400 600 1000
max_val
1000
min_val
200
minmax
num [1:5] 0 0.125 0.25 0.5 1
x
num [1:5] 10 20 30 40 50
Functions
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export

```
> data<-c(12,18,25,27,30,35,42,48,55,60,75,90,120)
> hist(data)
```